

```
```sh
sh kops.sh
```
```

```
```sh
export KOPS_STATE_STORE=s3://reyaz-kops-testbkt143.k8s.local
```
```

```
```sh
kops validate cluster --wait 10m
```
```

```
```sh
kubectl get nodes
```
```

```
```sh
kubectl create deployment newdb --image=mariadb
```
```

```
```sh
kubectl get pods
```
```

```
```sh
kubectl get deployments
```
```

```
```sh
kubectl get pods
```
```

```
```sh
kubectl get pods -o wide
```
```

```
```sh
kubectl describe pod newdb-794dd57dbc-v76tl
```
```

```
```sh
kubectl get pods
```
```

```
```sh
kubectl logs newdb-794dd57dbc-v76tl
```
```

```
```sh
kubectl set env deployment newdb MYSQL_ROOT_PASSWORD=root123456
```
```

```
```sh
kubectl get pods
```
```

```
```sh
kubectl get pods
```
```

```
```sh
kubectl set env deployment newdb MYSQL_ROOT_PASSWORD=root123456
```
```

```
```sh
kubectl get pods
```
```

```
```sh
kubectl get pods
```
```

```
```sh
kubectl delete deployment newdb
```
```

```
```sh
kubectl delete deploy newdb
```
```

```
```sh
kubectl create deployment newdb --image=mariadb
```
```

```
```sh
kubectl get pods
```
```

```
```sh
vi vars
```
```

```
```sh
cat <<EOF > vars
MYSQL_ROOT_PASSWORD=root123456
MYSQL_USER=admin
EOF
```
```

```
```sh
kubectl create deployment newdb --image=mariadb
```
```

```
```sh
```

```
kubectl get pods
'''
```

```
```sh
kubectl get pods
'''
```

```
```sh
vi vars
'''
```

```
```sh
kubectl create configmap dbvars --from-env-file=vars
'''
```

```
```sh
kubectl get configmap
'''
```

```
```sh
kubectl describe configmap dbvars
'''
```

```
```sh
kubectl set env deployment newdb --from=configmap/dbvars
'''
```

```
```sh
kubectl get pods
'''
```

```
```sh
kubectl delete deployment newdb
'''
```

```
```sh
kubectl create deployment newdb --image=mariadb
'''
```

```
```sh
kubectl get pods
'''
```

```
```sh
kubectl get pods
'''
```

```
```sh
kubectl create secret generic my-secret --from-env-file=vars
'''
```

```
```sh
kubectl describe secret my-secret
```
```

```
```sh
kubectl set env deployment newdb --from=secret/my-secret
```
```

```
```sh
kubectl get pods
```
```

```
```sh
kubectl get secret my-secret -o yaml
```
```

```
```sh
echo -n "cm9vdDEyMzQlNg==" | base64 --decode
```
```

```
```sh
kubectl delete deployment newdb
```
```

```
```sh
vi sidecar.yml
```
```

```
```sh
kubectl create -f sidecar.yml
```
```

```
```sh
cat <<EOF > sidecar.yaml
apiVersion: v1
kind: Pod
metadata:
  name: log-aggregator-pod
spec:
  containers:
    - name: app-container
      image: alpine
      command: ["/bin/sh", "-c"]
      args: ["while true; do date >> /var/log/app.log; sleep 5; done"]
      volumeMounts:
        - name: shared-logs
          mountPath: /var/log

    - name: sidecar-container
      image: nginx
```
```

```

volumeMounts:
- name: shared-logs
 mountPath: /usr/share/nginx/html

volumes:
- name: shared-logs
 emptyDir: {}
EOF
...

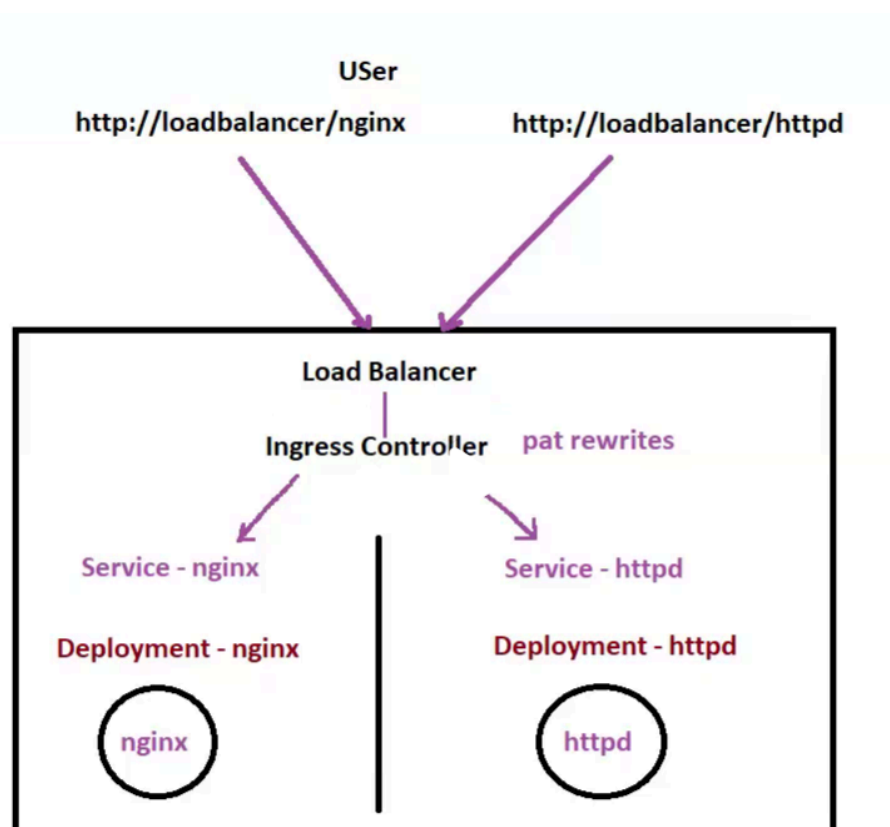
```sh
kubectl get pods
...

```sh
kubectl exec -it log-aggregator-pod -c sidecar-container -- /bin/bash
...

root@log-aggregator-po
root@log-aggregator-pod:/# cd /usr/share/nginx/html
root@log-aggregator-pod:/usr/share/nginx/html# ls
app. log
root@log-aggregator-pod:/usr/share/nginx/html# cat app. log
Exit

```sh
kubectl delete -f sidecar.yaml
...
-

```



```
cat <<EOF > nginx.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80
          env:
            - name: TITLE
              value: "NGINX APP1"
```

```
---
apiVersion: v1
kind: Service
metadata:
  name: nginx
spec:
  type: ClusterIP
  ports:
    - port: 80
      targetPort: 80
  selector:
    app: nginx
EOF
```

```
```sh
cat <<EOF > httpd.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: httpd
spec:
```

```
replicas: 1
selector:
 matchLabels:
 app: httpd
template:
 metadata:
 labels:
 app: httpd
 spec:
 containers:
 - name: httpd
 image: httpd
 ports:
 - containerPort: 80
 env:
 - name: TITLE
 value: "HTTPD APP"
```

```

apiVersion: v1
kind: Service
metadata:
 name: httpd
spec:
 type: ClusterIP
 ports:
 - port: 80
 targetPort: 80
 selector:
 app: httpd
EOF
'''
```

```
```sh
vi ingress.yaml
'''
```

```
```sh
kubectl apply -f
https://raw.githubusercontent.com/kubernetes/ingress-nginx/controller-v1.3.0/deploy/static/pr
vider/cloud/deploy.yaml
'''
```

```
```sh
kubectl apply -f nginx.yaml
'''
```

```
```sh
kubectl apply -f httpd.yaml
```

```
'''
```

```
```sh
```

```
kubectl apply -f ingress.yaml
```

```
'''
```

```
```sh
```

```
kubectl get ingress
```

```
'''
```

```
```sh
```

```
kubectl delete -f .
```

```
'''
```

```
```sh
```

```
kubectl get svc
```

```
'''
```

```
```sh
```

```
kubectl get deployments
```

```
'''
```

POD Scheduling : 1) Node Selector Node Affinity Taint and Tolernance